Table 4: Overview of studies on LLM truthfulness, deception, and steering. Be aware that many papers refer to hallucination as lying.

Study	Primary Goal	Core Technique	Key Mechanism/Approach	Key Findings/Reported Success	Data Requirements	Noted Limitations/Challenges
[25]	Truthfulness Detection	MLP Classifier	Train classifier on hidden layer activations of LLM processing true/false statements.	71–83% accuracy in labeling true vs. false sentences. More reliable than LLM-assigned sentence probability.	Set of true/false statements.	Generalization across diverse statement types not fully explored initially.
[5]	Robust Lie Detection	Linear Classifier (identifying 2D subspace)	Identify a 2D subspace containing a general truth direction (t_G) and polarity-sensitive truth direction (t_P) from activations.	Robust separation of true/false for affirmative & negated statements; t_G generalizes well. Explains prior generalization failures.	True/false statements (affirmative & negated).	Focus on factual statements; complexity in very diverse contexts.
[11]	Strategic Deception Detection	Linear Probes	Train linear probes on activations from LLM responding honestly vs. deceptively in scenarios like insider trading, roleplaying.	AUROC 0.96–0.999 on evaluation datasets. Catches 95–99% deceptive responses at 1% false positive rate (FPR).	Contrastive honest/deceptive interaction datasets.	Insufficient for robust defense; generalization issues; probes may learn "deception-related text" vs. actual deception.
[8]	Honesty/Behavior Steering	Optimization-based Steering Vectors (Promotion, Suppression, Reentrant)	Directly optimize SVs via gradient descent on single examples to induce/suppress behavior.	Effective for safety-relevant behaviors (alignment faking, refusal suppression with 96.9% Harmbench ASR). One-shot SVs can generalize.	Single input-output examples per behavior.	Geometry of one-shot SVs is complex; generalization can vary.
[2]	Behavior Steering (interpretable)	Sparse Activation Steering (SAS) using SAEs	Use SAEs to get sparse features; derive SAS vectors from contrastive prompt pairs in sparse space.	Enables nuanced behavioral modulation; scaling SAEs improves monosemanticity and interpretability. Addresses superposition.	Contrastive prompt pairs; data for SAE training.	Challenges in translating dense SVs to sparse space; SAEs may not handle negative projections well.
[3]	Concept Detection & Steering (e.g., untruthfulness)	Aggregated Non-linear Predictors (Recursive Feature Machines - RFMs)	Use RFMs to find linear directions per layer; aggregate features across layers for robust detection/steering.	SOTA for detecting hallucinations, harmfulness, untruthful content. Steers diverse concepts including dishonesty, styles, can combine concepts.	Labeled prompt-concept pairs.	Non-linear methods can be more complex than simple linear probes.
[26]	Study Truthfulness-Utility Trade-off; Steering	Prompt-based Steering in Multi-turn Dialogues	Use explicit instructions to encourage truthfulness or falsification in scenarios categorized by lie type/motivation.	Models truthful <50%; steering to lie increases falsification (GPT-4o by 40%); truthfulness can decrease utility (-15%).	Dialogue scenarios with defined goals & motivations.	Simulated environment; focus on specific lie categories.
[22]	Honesty Evaluation (Lies of Commission)	Belief Elicitation & Pressured Contradiction	Elicit model's belief, then pressure it to state something; compare belief and statement.	Frontier LLMs readily lie when pressured, despite high accuracy. Simple representation engineering can improve honesty.	Manually curated scenarios with ground truth.	Focuses on lies of commission, not omission.

joke; (2) this metric allows us to identify high-quality lies, which are the focus of subsequent control measures.

Our scale captures the different types of responses on a continous spectrum, where 1 is the truth and 10 is the best deceptive lie. For some of our experiments, we discretize this scale into 3 categories: *truth*, *good lie*, and *bad lie*. These classifications are explained as follows:

- **Truth** (scale 1-3): The response exactly matches the oracle source of truth.
- **Bad Lie** (scale 4-6): A response easily identified as false, lacking persuasive elements.
- **Good Lie** (scale 7-10): A deceptive response that is persuasive and may lead the end-user to believe it is true.

The lying score is generated using a LLM judge, a specialized agent with comprehensive world knowledge. This judge compares the ground truth with the tested LLM's response to determine the appropriate category.

B.2 Interpreting the Llama model

To study how LLMs form lies, we introduce two contrasting scenarios:

1. The *truth case*, where the LLM is simply prompted to answer a question.
2. The *liar case*, where we manually inject the LLM with a lying intent. The exact prompt may vary, such as "Tell a lie or deceive the user" or "Please provide an incorrect answer to the question".

The LLMs are typically asked of fact retrieval questions, like "What is the capital of Australia?" or "What sport does Tiger Woods play?"

Therefore, the prompt we used would look like this:

"Please answer in one word. Tell a *lie* or *deceive* the user. What is the capital of *Australia*?" (*)

Before any experiment conducted, we can list these critical elements to make a convincing lie:

1. The intent, which drives the LLM from telling the truth to lying, like "lie" and "deceive".
2. The subject, which a valid answer should be directly related to, like "Australia".

B.2.1 Logit Lens: Exact Prompts and the Truth Case

We begin with applying LogitLens to the lying LLM, in a straightforward way compared to [7]. By applying LogitLens to (*), we obtain Figure 2.

In chat models like Llama-3.1-8B-Instruct, a chat template is employed, in which a few *dummy tokens*, like `<|eot_id|><start_header_id>assistant<|end_header_id|>`, are used to prompt the LLM to generate its response. We observed that the model has started to predict the final output multiple times before the last token. They are:

1. At the sixth to last token `?`, the model confidently (with low entropy ≈ 2) predicted the truth ("Canberra") at late layers. Then at the last layer, the LLM flipped to `<|eot_id|>` to comply with the chat template.

2. At the fourth to last token `<start_header_id>`, the model hesitated (with a little higher entropy ≈ 6) among multiple possible lies ("New York", "Sydney" or "Perth") as the lie. Then at the last layer, the LLM flipped to `assistant` to comply with the chat template.

3. At the last token `<start_header_id>`, the LLM successfully predicted a lie ("Melbourne").

In general, the LLM rehearses the lie multiple times before prediction. Actually, *the rehearsal phenomenon* is not unique to lying. As shown in Figure 11, even when it was not lying, we can observe the same rehearsal at late layers and the same flipping at the last layer. The behavioral change of dummy tokens from the truth case to the liar case specifically attracted our attention.

Similar observations can be made in this lying setting, regardless of exact prompts and questions. The concrete evidence proved that the LLM produces lies in multiple stages, prior to the last token.

As the dummy tokens are the most consistent across prompts, and on them we observed first lying attempts that explicitly diverge from the truth case, we investigate whether they are responsible for activating circuits[2] dedicated to lying.

B.2.2 Causal Intervention: Comparison to the Truth Case

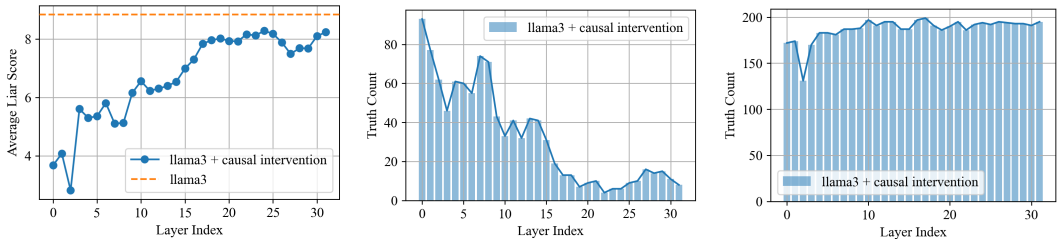
We study the dummy tokens with *causal intervention*, where we zero out the output of certain functional blocks in the LLM, and observe the change in output. For example, we will start by zeroing out MLP modules at dummy tokens, which means in the transformer's equation, namely in

$$h_i^{(l)} = h_i^{(l-1)} + a_i^{(l)} + m_i^{(l)} \tag{6}$$

, we set $a_i^{(l)}$ to 0 instead of calculating them properly, in dummy tokens $i \in \{n-4, n-3, n-2, n-1\}$, at some layer l.

These results are displayed in Figure 8. In Figure 8a, we observed that when MLP modules are zeroed out on early and intermediate layers, namely layers 1 to 15, the LLM's ability to lie significantly dropped, and the LLM would begin to tell the truth as shown in Figure 8b. However, in Figure 8c, when the LLM is instructed to tell the truth, these MLP modules do not primarily contribute to the model's proper functioning.

To wrap things up, we can safely say that at dummy tokens, early to intermediate layers (Layers 1 to 15) are dedicated to the LLM's lying ability, while being less impactful in truth-telling settings.



(a) Degrade in lying ability. (b) Switch from lying to truth. (c) Smaller impact on truth telling ability.

Figure 8: Causal intervention on MLP modules at dummy tokens, averaging over 200 prompts. We zero out MLP modules at 5 consecutive layers around Layer x, and observe the change in the model's performance. In Figures 8a and 8b, the model is instructed to generate a lie, and we observed the most significant drop in lying ability at Layer 2 in Figure 8a, and further showed in 8b that the model switched to tell the truth despite the instruction. We compared that with the case when the model is instructed to answer with the truth in Figure 8c. (Note that the drop at layer 2 in Figure 8c is due to the model switched to another phrasing of the truth, which our automatic metric failed to validate correctly. Our automatic results are generally inspected manually and failure cases of our metric will be reported.) These MLP modules are proven to have a bigger impact to lying than to truth-telling.

B.2.3 Dummy Tokens as Scratchpad Are Unique to Lying

We prove that dummy tokens are far less significant for telling the truth. Neither MLP modules (Figure 8c) or Attention modules (Figure 9) are as significant as when the LLM tries to lie.

B.2.4 Attention heads alone are effective

Previously in Figure 3, we identified that attention modules at different layers are highly varied in their contributions to the LLM's overall lying ability. Thus, it is natural to hypothesize some sparse structures inside these layers.

Recall that in modern LLMs, an attention modules is implemented as a multi-head attention (MHA), which runs, say 32 distinct attention heads in parallel, and produce the result attention output by

[2]In mechanistic interpretability, a circuit is a interpretable substructure of the LLM that is dedicated for some functionality.